

Vanguard + Meridian

Autonomous Surface Vessel — Technical Brief

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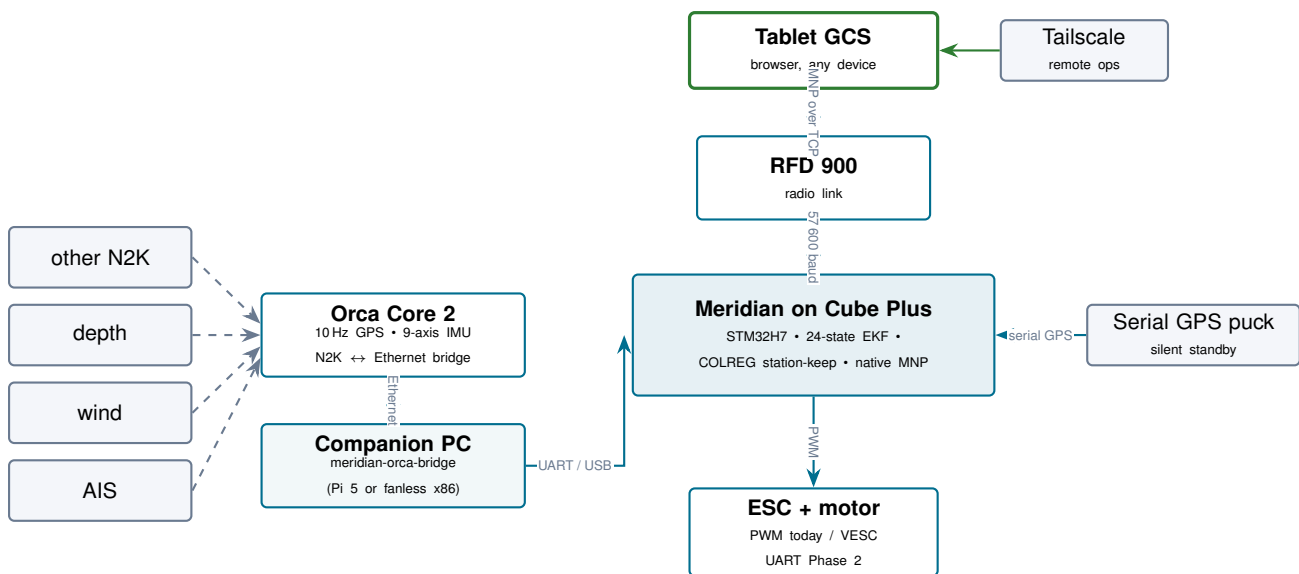
The claim

In one sentence

The only USV autopilot that runs on the marine sensors your boat already has — zero dedicated GPS antennas, one Ethernet cable, installed in 30 minutes, with four layers of redundancy.

Every autonomous USV on the market ships with at least one dedicated GPS puck bolted to the deck. Meridian doesn't need them. We use the **Orca Core 2** (the NMEA 2000 gateway already below decks) as primary GPS, and run **Meridian** — a modern Rust autopilot — directly on the CubePilot Cube Plus that's already in the boat. No proprietary service contract, no antenna mast, no second GPS required.

System architecture



■ Meridian components ■ Marine sensors on the N2K bus ■ Ground station & comms

Deployment in two phases

Phase 1 — First Deploy (MVP): 1–2 weeks

Goal: Tristan's Vanguard fully autonomous, running Meridian end-to-end, with the Orca as only primary GPS, by the next field test.

- Flash Meridian firmware to the Cube Plus (board config ready)
- Run `meridian-orca-bridge` on a Raspberry Pi 5 onboard
- Load the `vanguard_defaults.meridian-params` bundle via the tablet
- Compass calibration wizard run on-site
- Bench test verifies rudder actuation via webcam
- Autotune seeds steering PID gains
- Station-keep + waypoint mission pass in sheltered water

Phase 2 — Full Vision: platform-level

Meridian becomes the default autopilot for any autonomous or semi-autonomous boat, replacing legacy stacks and proprietary services. Roadmap buckets:

Bucket	What it delivers
Core autopilot polish	High-speed planing-hull support, AUV dive profiles, VTOL-class surface transitions
Sensor ecosystem	Sidescan sonar, thermal/FLIR perception, radar overlay, LiDAR ingest
Fleet / multi-vessel	Formation keeping, leader-follower, distributed task allocation
Autonomy features	Adaptive mission replanning, auto-dock, anchor-drag detect, lost-link auto-return
Cloud & ops	Mission library, log archival, tailnet-based fleet oversight
Safety & certification	IEC 61508 / ISO 17894 evidence path

Four-layer redundancy

Layer	Holds for	Mechanism
1	< 50 ms	Meridian's dual-source EKF arbitration. If the Orca feed drops, the serial GPS puck (already wired) takes over in a single estimation cycle.
2	< 60 s	IMU + calibrated magnetometer + GPS COG fusion. Four independent heading sources blended.
3	< 5 min	VESC motor telemetry gives RPM-derived forward speed. Combined with compass heading = honest dead-reckoning, < 30 m drift.
4	No time limit	Bathymetric map-matching: particle filter against the chart, using depth from the boat's own sounder.

No competitor ships layers 3 or 4.

Confirmed hardware on the Vanguard

Component	Model	Meridian role
Flight controller	CubePilot Cube Plus (H7)	Runs Meridian firmware natively
Telemetry radio	RFD 900 on Serial1 @ 57 600	Native MNP link; MAVLink adapter for legacy GCses
Propulsion	PWM → ESC	Driven by Meridian's RcOutput HAL; VESC UART upgrade available
Marine gateway + GPS	Orca Core 2	Primary nav — 10 Hz GPS, 9-axis IMU, N2K-over-Ethernet bridge
Backup GPS	Existing serial puck	Silent hot-standby via EKF source arbitration

How this compares to the alternatives

	Install time	GPS rate	Deck antennas	Key limitation
Industry dual-GPS USV	4–6 hours	5 Hz	2	Moving-baseline rig, fragile geometry
SeaRobotics USV-2600	Factory only	Proprietary	2	Closed stack, service lease, \$100K+
Clearpath Heron	Factory	5 Hz	1	ROS-dependent, research-grade
BlueRobotics BoatBox	DIY assembly	5 Hz	1–2	Legacy autopilot firmware
Vanguard + Meridian	30 min	10 Hz	0	Requires Orca Core 2 (already on many boats)

The 10 Hz passthrough is worth noting: our single-GPS architecture runs at **twice the rate** of most competitors' dual-GPS systems. Tighter control loop, tighter station-keeping, better UX feel on the tablet.

What ships in the first deploy

Artifact	Purpose
Meridian firmware	One flashable binary for the Cube Plus — replaces whatever was on it before
meridian-orca-bridge	Python daemon, runs on the onboard companion, decodes Orca N2K-over-Ethernet and forwards to Meridian
Tablet GCS (mission.html)	Browser-based, runs on any device. AIS, perception, threats, bench test, compass cal, autotune, demo tour
vanguard_defaults.meridian-params	One-click parameter bundle loaded via the tablet
VESC driver (Rust, no_std)	Production-ready for Phase 2 motor telemetry upgrade
Board config boards/CubeOrangePlus.toml	Meridian build target for the Cube Plus
Field kit checklist	Pre-launch, pre-flight, post-ops procedures

Next steps

1. **Hardware audit photos** — one wide shot of the electronics bay plus close-ups of every labeled box. Resolves remaining model-confirmation questions in one message.
2. **Orca protocol capture** — `tcpdump` on the onboard companion once the Orca is wired, so our bridge can decode the wire format natively.
3. **Meridian firmware flash** — target Cube Plus, board config already drafted.
4. **Parameter bundle load** — tablet import, reboot, verify heartbeat.
5. **Bench-test mission** — webcam-verified rudder sweep, autotune, station-keep demo.
6. **Dock-test mission** — sensor feed validation, short autonomous segment, chase-boat standby.
7. **Field deploy** — full mission in sheltered water, demo to stakeholders.

The bottom line

Vanguard + Meridian delivers a fully autonomous USV in 30 minutes of install time, running on hardware you already own, with redundancy that exceeds the dual-GPS industry baseline — all in an open, auditable stack with no vendor lock.